

Stat 437 HW4A

Your Name (Your student ID)

General rule

Please show your work and submit your computer codes in order to get points. Providing correct answers without supporting details does not receive full credits. This HW covers

- Discriminant analysis

For an assignment or project, you DO NOT have to submit your answers or reports using typesetting software. However, your answers must be well organized and well legible for grading. Please upload your answers in a document to the course space. Specifically, if you are not able to knit a .Rmd/.rmd file into an output file such as a .pdf, .doc, .docx or .html file that contains your codes, outputs from your codes, your interpretations on the outputs, and your answers in text (possibly with math expressions), please organize your codes, their outputs and your answers in a document in the format given below:

Problem or task or question ...

Codes ...

Outputs ...

Your interpretations ...

It is absolutely not OK to just submit your codes only. This will result in a considerable loss of points on your assignments or projects.

Conceptual exercises: III (Discriminant analysis)

3. Exercise 2 of Section 4.7 of the Text, which starts with “It was stated in the text that classifying an observation to the class for which (4.12) is largest is equivalent to classifying an observation to the class for which (4.13) is largest. Prove that this is the case.” (Helpful information on how to prove this is contained in the lecture video on LDA and “LectureNotes5b_notes.pdf”.)

4. Exercise 3 of Section 4.7 of the Text, which starts with “This problem relates to the QDA model, in which the observations within each class are drawn from a normal distribution with a class specific mean vector and a class specific covariance matrix. We consider the simple case where $p = 1$; i.e. there is only one feature.” (Helpful information on how to prove this is contained in the lecture video on QDA and “LectureNotes5b_notes.pdf”.)

5. Exercise 5 of Section 4.7 of the Text, which starts with “We now examine the differences between LDA and QDA.” (Hint: for this question, you may also use information from Figures 4.9, 4.10 and 4.11 in the Text.)

6. Let Y be the random variable for the class label of a random vector $X \in \mathbb{R}^p$ (where p is the number of features), such that $Y \in \mathcal{G} = \{1, \dots, K\}$ and $\Pr(Y = k) = \pi_k$ for Class k with $k \in \mathcal{G}$, where $K \geq 2$ is the number of classes. Consider the Gaussian mixture model such that the conditional density of X when it comes from Class k is $f_k(x) \sim \text{Gaussian}(\mu_k, \Sigma_k)$. Given the training set $\mathcal{T}$ of

n observations $(x_1, y_1), \dots, (x_n, y_n)$ on (X, Y) , where y_i is the class label of observation x_i , do the following:

6.1) Provide the MLEs of π_k , μ_k and Σ_k for each $k \in \mathcal{G}$ respectively for the case where all Σ_k 's are equal and for the case where not all Σ_k 's are equal. When $p > n$, is the MLE of Σ_k still accurate? If not, recommend a different estimator for estimating Σ_k and provide details on this estimator.

6.2) Assume $p = 2$ and $K = 2$ and $k = 1$. For the density $f_k(x) \sim \text{Gaussian}(\mu_k, \Sigma_k)$, what shape do its contours take, and how does Σ_k control the shape of these contours? How do you check if the conditional density of X given that it comes from Class k is Gaussian?

6.3) Is it true that discriminant analysis will perform badly if the Gaussian assumption is violated? (Hint: for this question, you may also use the information provided by Figures 4.10 and 4.11 of the Text.) Let $X = (X_1, \dots, X_p)^P$, i.e., X_1 up to X_p are the feature variables. Can discriminant analysis be applied to observations of X when some of $X_j, j = 1 \dots, p$ is a discrete variable (such as a categorical variable)? Explain your answer.

6.4) What is a ROC curve, and what is AUC? How is AUC used to gauge the performance of a classifier? If you apply the same classifier, say, LDA or QDA under the same Gaussian mixture model, to two data sets that are independently generated from the same data generating process, i.e., that are independently generated from (X, Y) for classification problems, and obtain two ROC curves, would the two ROC curves be quite different? Explain your answer. When there are 3 or more classes, are the codes provided in the lecture notes able to obtain ROC curves and their AUC's for LDA and QDA?

6.5) Describe the key similarities and differences, respectively, between LDA and logistic regression. Provide a situation where discriminant analysis can still be sensibly applied but logistic regression is not well-defined.

Applied exercises: III (Discriminant analysis)

8. The following is on software commands:

(8.1) What is the main cause of the message “Warning in `lda.default(x, grouping, ...)`: variables are collinear”? What is the main cause of the message “Error in `qda.default(x, grouping, ...)` : some group is too small for ‘qda’”?

(8.2) Provide details on the `list` that `predict{MASS}` returns.

(8.3) The arguments `gamma` and `lambda` of `rda{klaR}` are usually determined by cross-validation. Can they be set manually?

9. We will use the human cancer microarray data that were discussed in the lectures and are provided by the R library `ElemStatLearn` (available at <https://cran.r-project.org/src/contrib/Archive/ElemStatLearn/>). Pick 3 cancer types “MELANOMA”, “OVARIAN” and “RENAL”, and randomly select the same set of 60 genes for each cancer type. Please use `set.seed(123)` for the whole of this exercise. Your analysis will be based on observations for these genes and cancer types.

9.1) Pick 2 features and visualize the observations using the 2 features. Do you think it is hard to classify the observations based on the amount of overlap among the 3 neighborhoods of observations

in each of the 3 classes? Here “a neighborhood of observations in a class” is a “open disk that contains the observations in the class”.

9.2) Apply LDA and report the classwise error rate for each cancer type.

9.3) Use the library `klaR`, and apply regularized discriminant analysis (RDA) by setting the arguments `gamma` and `lambda` of `rda{klaR}` manually so that the resulting classwise error rate for each cancer type is zero.

9.4) Obtain the estimated covariance matrices from the RDA and visualize them using the same strategy in Example 3 in “LectureNotes5c_notes.pdf”. What can you say about the degree of dependence among these genes for each of the three cancer types? (Hint and caution: the class labels “MELANOMA”, “OVARIAN” and “RENAL” will be ordered alphabetically by R. So, you need to keep track on which estimated covariance matrix is for which class. Otherwise, you will get wrong visualization.)